MCU1 / J1 / W5500(A1) 接続一覧 / MCU1 / J1 / W5500 (A1) Connection List

本書は日本語と英語を併記している。各節で日本語の後に英語が続く。表は各欄に日本語と英語を並べて書く。

This document is written in both Japanese and English. In each section, the Japanese text is followed by the English text. In the tables, each cell gives the Japanese and then the English.

KiCad回路図(`hardware/W5500_PA_PTT_Control/kicad/w5500-esp32.kicad_sch`)のnetlistから抽出した、MCU1(ESP32 DevKitC、参考表示のみ・DNP)・J1(実体のDevKitC取付位置。ピンソケットは使わず直接はんだ付けする)・A1(W5500_Lite)間の対応表。

MCU1は回路図上の参考表示シンボルでDNP(実装しない)。実体はJ1に直接はんだ付けするFreenove DevKitC基板のため、SPI信号(CS/SCK/MISO/MOSI)はMCU1側では別ネット名(ラベルのみ)であり、実際の結線はJ1-A1間の直接ワイヤ。

J1は2026-08-30に実機(Freenove ESP32-WROOM-32E DevKitC)の実ピン配列に合わせて2x19(38ピン)から2x20(40ピン)へ修正した。さらに2026-08-31、MCU1もJ1と同じ`project_symbols:ESP32_DevKitC_Socket_40P`シンボルに置き換え、**MCU1のピン番号・信号名はJ1と完全に一致**するようになった（従来はMCU1がESP32-WROOM-32モジュール自体のパッド番号(KiCad標準`RF_Module:ESP32-WROOM-32`ライブラリ)を使っていたため、J1のピン番号と一致しなかった）。以下の表のピン番号はMCU1・J1共通。

A mapping table between MCU1 (ESP32 DevKitC, reference display only, DNP), J1 (the actual DevKitC mounting position; soldered directly, without a pin socket) and A1 (W5500_Lite), extracted from the netlist of the KiCad schematic (`hardware/W5500_PA_PTT_Control/`

kicad/w5500-esp32.kicad_sch).

MCU1 is a reference-display symbol on the schematic and is DNP (not populated). The actual part is the Freenove DevKitC board soldered directly to J1, so the SPI signals (CS/SCK/MISO/MOSI) have separate net names (labels only) on the MCU1 side, and the actual wiring is direct wires between J1 and A1.

On 2026-08-30, J1 was corrected from 2x19 (38 pins) to 2x20 (40 pins) to match the actual pinout of the real board (Freenove ESP32-WROOM-32E DevKitC). Furthermore, on 2026-08-31 MCU1 was also replaced with the same `project_symbols:ESP32_DevKitC_Socket_40P` symbol as J1, so **MCU1's pin numbers and signal names now match J1 exactly** (previously MCU1 used the pad numbers of the ESP32-WROOM-32 module itself (KiCad standard `RF_Module:ESP32-WROOM-32` library), which did not match J1's pin numbers). The pin numbers in the table below are common to MCU1 and J1.

2026-09-30、フットプリントを実物の資料に合わせて2点修正した。- A1(W5500 Lite)は、USR-ES1データシート(Jinan USR IOT「3.3 Dimensions」)に合わせて、ピン列の間隔を15.24mmから20.32mmへ、7〜12番の並びを1〜6番と同じくRJ45側から数える順に修正した(従来は7番がRJ45と反対側で、右列の信号が逆につながっていた)。フットプリント名は`W5500_Lite_USR-ES1_2x06_P2.54mm_20.32mm`。- J1(DevKitC取付位置)は、Freenove公式のピン配置図(部品面から見た図)に合わせて、ピンの並びを上下反転した。1番(5V)・21番(GND)がUSB-C側、20番(3V3)・40番(GND)がアンテナ側になる(従来は鏡像の並びで、5Vが3V3やGNDのピンにつながる配置だった)。

On 2026-09-30, two footprints were corrected to match the documentation of the actual parts. - A1 (W5500 Lite): per the USR-ES1 datasheet (Jinan USR IOT, "3.3 Dimensions"), the row spacing was changed from 15.24 mm to 20.32 mm, and pins 7–12 now count from the RJ45 end like pins 1–6 (previously pin 7 was at the end opposite the RJ45, so the right-row signals were connected in

reverse). The footprint name is `W5500_Lite_USR-ES1_2x06_P2.54mm_20.32mm`. - J1 (DevKitC mounting position): per Freenove's official pinout diagram (seen from the component side), the pin order was flipped vertically. Pins 1 (5V) and 21 (GND) are at the USB-C end and pins 20 (3V3) and 40 (GND) at the antenna end (previously the order was a mirror image, placing 5V on the 3V3 or GND pins).

信号 Signal	MCU1/J1ピン MCU1/J1 pin	W5500(A1)ピン W5500 (A1) pin	備考 Notes
SPI CS	GPIO 5 (31)	CS (5)	MCU1側は別ネット名(ラベルのみ)、実配線はJ1-A1間 Separate net name on the MCU1 side (label only); actual wiring is between J1 and A1
SPI SCK	GPIO 18 (32)	SCK (4)	同上 Same as above
SPI MISO	GPIO 19 (33)	MISO (12)	同上 Same as above
SPI MOSI	GPIO 23 (39)	MOSI (3)	同上 Same as above
W5500 RST	GPIO 21 (35)	RST (11)	3者同一ネット(GPIO制御リセット。ファームウェアで起動時にパルス送出)。R9(10kΩ)で+3V3_Aへプルアップ Same net for all three (GPIO-controlled reset; the firmware sends a pulse at startup). Pulled up to +3V3_A by R9 (10 kΩ)

POWER(12V電源)制御 POWER (12 V power) control	GPIO 26 (11)	—	W5500非接続。ネットラベルはGPIO26_PAだが、実体は12V電源のON/OFF制御（下表参照） Not connected to the W5500. The net label is GPIO26_PA, but it actually controls the 12 V power ON/OFF (see the table below)
PTT制御 PTT control	GPIO 27 (10)	—	W5500非接続。ネットラベルはGPIO27_PTT。無線機PTT端子をGND側へ落とす方式（下表参照） Not connected to the W5500. The net label is GPIO27_PTT. Pulls the radio's PTT terminal to GND (see the table below)
GND	GND (7, 21〜24, 40)	1, 2, 7	共通GND。J1は7・21〜24・40番をGNDに接続(34番のみ未結線。ボード内部でつながっている) Common GND. On J1, pins 7, 21–24 and 40 are connected to GND (only pin 34 is left unconnected; it is connected inside the board)
+5V(+5v0、DevKitC基板給電) +5 V (+5v0, DevKitC board supply)	5V (1, 2)	—	U2(L7805、+12V→+5V)から供給。MCU1(ESP32モジュール自体)は3.3V駆動のため対応ピンなし。DevKitC基板上のオンボードLDOがこの5VをESP32モジュール用3.3Vへ変換する Supplied from U2 (L7805, +12 V → +5 V). MCU1 (the ESP32 module itself) runs on 3.3 V and has no corresponding pin. The on-board LDO of the DevKitC board converts this 5 V to 3.3 V for the ESP32 module

+3.3V(ESP32系) +3.3 V (ESP32 side)	3.3V (3)	—	DevKitC基板のオンボードLDO出力(上記+5Vから生成)。MCU1(ESP32モジュール)のVDDピンへの供給元。J1・MCU1とも他に3.3Vピンが4,5,20にあるが未結線 Output of the DevKitC board's on-board LDO (generated from the +5 V above). The supply for the VDD pin of MCU1 (the ESP32 module). J1 and MCU1 also have 3.3 V pins at 4, 5 and 20, which are unconnected
+3V3_A(W5500専用) +3V3_A (W5500 only)	—	VCC (8, 9)	上記と同じ+5v0(U2の出力)をU1(LM2940T-3.3、+5V→+3.3V)が受けて生成。ESP32側3.3V(DevKitC基板内蔵LDO経由)とは生成回路が別だが、元の5Vは共通 Generated by U1 (LM2940T-3.3, +5 V → +3.3 V) from the same +5v0 (U2 output) as above. The generating circuit is separate from the ESP32 side's 3.3 V (via the DevKitC board's built-in LDO), but the source 5 V is common

未使用: A1のINT(6)・NC(10)、J1・MCU1の残りGPIOピン(GPIO13,GPIO12,GPIO14,GPIO33,GPIO32,GPIO35,GPIO34,VN,VP,EN,GPIO15,GPIO2,GPIO0,GPIO4,GPIO16,GPIO17,RX,TX,GPIO22)は今回未接続。

Unused: A1's INT (6) and NC (10), and the remaining GPIO pins of J1/MCU1 (GPIO13, GPIO12, GPIO14, GPIO33, GPIO32, GPIO35, GPIO34, VN, VP, EN, GPIO15, GPIO2, GPIO0, GPIO4, GPIO16, GPIO17, RX, TX, GPIO22) are not connected this time.

MCU1シンボルは回路図上「参考表示のみ・DNP」であり、実体はJ1に直接はんだ付けするFreenove DevKitC基板そのものを表す。2026-08-31以降はJ1と同じproject_symbols: ESP32_DevKitC_Socket_40Pシンボルを使っており、DevKitC基板自体には無いESP32モジュール固有のフラッシュ用ピン(SD0〜SD3/CMD/CLK等)は表示されない。

ESP32モジュールチップ自体は3.3V駆動で5V入力ピンを持たないため、

+5V系統にMCU1側の対応ピンは存在しない……という制約は旧シンボルの話であり、現行のDevKitCシンボル表示では「5V」ピンも(実際にDevKitC基板が受電する電源として)表示されるが、本回路では未結線のまま。実際に+5Vを受電するのはJ1にはんだ付けするFreenove DevKitC基板側であり、基板上のオンボードLDOでの5V→3.3V変換はこの回路図の範囲外（DevKitC基板内部）である。

W5500(A1)用の+3V3_Aは、MCU1(DevKitC基板)用の+5Vと**同じ`+5v0`ネット**（U2=L7805の出力）をU1(LM2940T-3.3)がさらに3.3Vへ降圧して生成する。すなわち12V→5Vへの降圧はU2(L7805)1系統のみで、そこからMCU1用とW5500用の2方向に分岐する構成（外付けDCDCバックコンバータモジュールは使用しない）。

The MCU1 symbol is "reference display only, DNP" on the schematic and represents the Freenove DevKitC board itself soldered directly to J1. Since 2026-08-31 it uses the same `project_symbols:ESP32_DevKitC_Socket_40P` symbol as J1, so the ESP32-module-specific flash pins (SD0–SD3/CMD/CLK, etc.) that the DevKitC board itself does not have are not shown.

The constraint that "the ESP32 module chip itself runs on 3.3 V and has no 5 V input pin, so there is no corresponding MCU1 pin for the +5 V system" applied to the old symbol. With the current DevKitC symbol display, the "5V" pin is also shown (as the power the DevKitC board actually receives), but it remains unconnected in this circuit. What actually receives the +5 V is the Freenove DevKitC board soldered to J1, and the 5 V → 3.3 V conversion by its on-board LDO is outside the scope of this schematic (inside the DevKitC board).

The +3V3_A for the W5500 (A1) is generated by U1 (LM2940T-3.3) further stepping down to 3.3 V **the same `+5v0` net** (U2 = L7805 output) as the +5 V for MCU1 (DevKitC board). In other words, the 12 V → 5 V step-down is a single system through U2 (L7805), which branches in two directions from there, for MCU1 and for the W5500 (no external DC-DC buck converter module is used).

GPIO25(J1・MCU1ピン12)はRev.1.1時点でLNA制御用ネット(GPIO25_LNA)としてラベル付けされていたが、本表・下表いずれの駆動回路にも接続されておらず、現行回路では未使用（PA_Power/PTTコントローラの対象外）。2026-10-01、回路図に残っていたGPIO25_LNAラベルを削除し、J1・MCU1の12番ピンを未接続(×)にした。

GPIO25 (J1/MCU1 pin 12) was labeled as the LNA control net (GPIO25_LNA) as of Rev.1.1, but it is not connected to any drive circuit in this table or the table below and is unused in the current circuit (outside the scope of the PA_Power/PTT controller). On 2026-10-01 the leftover GPIO25_LNA label was removed from the schematic, and pin 12 of J1 and MCU1 was marked as no-connect (×).

POWER(12V電源)・PTT 駆動回路の接続詳細 / POWER (12 V Power) and PTT Drive Circuit Connection Details

GPIO26・GPIO27はW5500とは独立に、以下の駆動回路経由でJ5(Power)・J6(PTT_ON)へ接続される。

GPIO26 and GPIO27 are connected to J5 (Power) and J6 (PTT_ON) through the following drive circuits, independently of the W5500.

GPIO	経路 Path	部品 Parts	備考 Notes

GPIO26 (POWER)	GPIO26 → R6 → Q1(B)、R12でGNDへプルダウン GPIO26 → R6 → Q1 (B), pulled down to GND by R12	R6: 510Ω、R12: 10kΩ、Q1: 2SC1815(NPN) R6: 510 Ω, R12: 10 kΩ, Q1: 2SC1815 (NPN)	GPIO26 HIGHでQ1がONし、Q5のゲートをGND側へ引く。GPIO26が確定しない間はR12でQ1をOFFに保つ When GPIO26 is HIGH, Q1 turns ON and pulls Q5's gate toward GND. While GPIO26 is undetermined, R12 keeps Q1 OFF
↳	Q1(C) → Q5(G)、R8でプルアップ Q1 (C) → Q5 (G), pulled up by R8	R8: 1kΩ(+12Vへプルアップ、1/2W)、Q5: 2SJ334(PMOS, TO-220、足は1=G・2=D・3=S、放熱器付き) R8: 1 kΩ (pull-up to +12 V, 1/2 W), Q5: 2SJ334 (PMOS, TO-220, pins 1 = G, 2 = D, 3 = S, with heatsink)	Q1 OFF時はR8でゲートが+12Vにプルアップされ、Q5(ハイサイドスイッチ)はOFF When Q1 is OFF, R8 pulls the gate up to +12 V and Q5 (high-side switch) is OFF
↳	Q5(S)=+12V、Q5(D) → J5(Power) Q5 (S) = +12 V, Q5 (D) → J5 (Power)	J5: Power出力コネクタ J5: Power output connector	Q5 ONで+12VがJ5(Power)へ導通（旧リレーK3の置き換え） When Q5 is ON, +12 V conducts to J5 (Power) (replaces the former relay K3)

GPIO27 (PTT)	GPIO27 → R7 → Q3(B)、R13でGNDへプルダウン GPIO27 → R7 → Q3 (B), pulled down to GND by R13	R7: 510Ω、R13: 10kΩ、Q3: 2SC1815(NPN) R7: 510 Ω, R13: 10 kΩ, Q3: 2SC1815 (NPN)	GPIO27 HIGHでQ3がONする。GPIO27が確定しない間はR13でQ3をOFFに保つ When GPIO27 is HIGH, Q3 turns ON. R13 keeps Q3 OFF while GPIO27 is undetermined
↳	Q3(C) → J6(PTT_ON) Q3 (C) → J6 (PTT_ON)	—	Q3(オープンコレクタ的動作)ONでJ6(PTT_ON)をGND側へ落とす（無線機PTT端子への接続を想定。Q3 OFF時はハイインピーダンス） When Q3 (acting like an open collector) is ON, it pulls J6 (PTT_ON) to GND (intended for connection to the radio's PTT terminal; high impedance when Q3 is OFF)

Q1・Q3のエミッタ(pin1)は共通GNDに接続。

The emitters (pin 1) of Q1 and Q3 are connected to the common GND.

電源系統の接続詳細 / Power System Connection Details

外部から供給される+12V(13.8V)を起点に、**U2(L7805)が生成する+5V(+5v0)をMCU1(DevKitC基板)とU1(LM2940T-3.3)の両方に分岐供給**する。12V→5Vへの降圧はU2の1系統のみで、外付けのDCDCバックコンバータモジュールは使用しない。

Starting from the externally supplied +12 V (13.8 V), **the +5 V**

(+5v0) generated by U2 (L7805) is branched to both MCU1 (DevKitC board) and U1 (LM2940T-3.3). The 12 V → 5 V step-down is a single system through U2, and no external DC-DC buck converter module is used.

経路 Path	部品 Parts	備考 Notes
電源入力: J2(DC_IN_13V8) → +12Vネット Power input: J2 (DC_IN_13V8) → +12 V net	J2: 2ピン (+12V, GND) J2: 2 pins (+12 V, GND)	外部電源(13.8V)の入力コネクタ Input connector for the external power supply (13.8 V)
+12V → U2(IN) → U2(OUT) =+5v0	U2: L7805(TO-220、12V→5V)	C3(0.33μF、25V以上)が+12V-GND間のバイパス C3 (0.33 μF, 25 V or higher) bypasses +12 V to GND
+5v0 → J1(1,2)	—	MCU1(DevKitC基板)へ供給。基板上のオンボードLDOで3.3Vに変換されESP32モジュールへ渡る Supplies MCU1 (DevKitC board). Converted to 3.3 V by the on-board LDO and passed to the ESP32 module

+5v0 → U1(IN) → U1(OUT) =+3V3_A → A1(VCC)	U1: LM2940T-3.3(TO-220、5V→3.3V)	C1(0.47μF)が+5v0-GND間のバイパス(LM2940の入力コンデンサ)、C2(33μF電解、10V以上)が+3V3_A-GND間の出力コンデンサ(LM2940の発振防止に必要。22μF以上・ESR 0.1〜1Ω) C1 (0.47 μF) bypasses +5v0 to GND (the LM2940 input capacitor), and C2 (33 μF electrolytic, 10 V or higher) is the output capacitor on +3V3_A (needed to keep the LM2940 from oscillating: at least 22 μF, ESR 0.1–1 Ω)

+5v0(U2の出力)は、J1(MCU1用)とU1(W5500用3.3Vレギュレータの入力)の**共通の供給源**であり、この2つは同一ネットで並列に分岐している点に注意（Rev.2.0時点では外付けDCDCバックコンバータ用のJ3が別系統として存在したが、2026-08-30のRev.2.1で廃止しU2からの単一系統に統合した）。

Note that +5v0 (U2 output) is the **common supply** for J1 (for MCU1) and U1 (input of the 3.3 V regulator for the W5500), and the two branch in parallel from the same net (as of Rev.2.0, J3 for an external DC-DC buck converter existed as a separate system, but it was removed in Rev.2.1 on 2026-08-30 and merged into the single system from U2).

関連ファイル / Related Files

- 回路図 / Schematic: `hardware/W5500_PA_PTT_Control/kicad/w5500-esp32.kicad_sch`
- SPI配線の簡易表(ESP32 GPIO番号のみ) / Simple SPI wiring table (ESP32 GPIO numbers only): `W5500_PA_PTT_Control_仕様書.md` 2.2節 / section 2.2
- ファームウェア / Firmware: `hardware/W5500_PA_PTT_Control/W5500_PA_PTT_Control.ino`